MFE FAQ

- Before you trigger a build
- General
  - How can I enable logging of components
  - New MFEs created with React 19 incompatible with existing React 18 SPA/MFE
  - How to load hosted widgets in local MFE
- Build Errors
  - 403 Not Enough permissions to delete/overwrite artifact
  - withOptions is not a function
  - npm ERR! Idle timeout reached for host artifactory.srv.westpac.com.au and "cache miss" of packages
  - HookWebpackError: ENOENT: no such file or directory
- Deploy Errors
  - K8s Delpoy - No scopes defined ....
  - K8s Deploy - Could not promote exp/docker
  - K8s Deploy - Could not deploy exp/docker
  - Internal GW - Upsert Swagger File Stage
  - Deploy UI Stage
  - PIV UI Deployment Stage
- Runtime Errors
  - Application not accessible / SPA not found
  - Uncaught (in promise) Error: Loading chunk ...
  - Application loader keeps on loading and routes disallowed
  - 503 error returned from BFF API
  - 504 error returned from gateway
  - 500 error timeout returned from BFF API
  - High CPU observed in containers
  - healthcheck returns response code 0

Before you trigger a build

Most pipeline build errors can be easily avoided by following these steps on your local machine, before triggering a pipeline build:

1. delete all node_modules (optional if you know your dependency tree is in sync with package-lock.json)
2. npm ci
3. npm run build
4. npm run test
5. npm start

Resolve any errors above before proceeding to run a build on the build pipeline.

General

How can I enable logging of components

Logging to splunk

Is sole responsibility of federated development teams to ensure logs are set at an appropriate log level in their source code. Platform provides a deploy flag to enable development teams to override the platform default levels. Refer to Customise BFF or EXP API log levels

Platform default log level

DEV1	DEBUG
SIT1	DEBUG
UAT1	INFO
SVP1	INFO
PROD	INFO

Logging of Mesh framework components

Used only in non production, to assist in debug of platform managed framework components

Setup logging of node-framework components - Westpac Mesh - Westpac Enterprise Confluence

New MFEs created with React 19 incompatible with existing React 18 SPA/MFE

Presently Mesh UI create SPA/MFE components with the latest stable version of React (19)

We are aware of issues at runtime when any other component in your SPA/MFE/widget stack is running a different version of React.

Solution

Please use only one version of React (18 or 19) for your entire SPA/MFE stack, and use the version that matches the lowest version in your UI component stack. ie. If necessary please downgrade any new MFE that was created in React 19 to match the lowest version (SPA/MFE) in your application.

Applications that only have React 19 components, no action is necessary.

How to load hosted widgets in local MFE

When you are developing MFE locally, and need to reference an existing hosted widget without running the widget locally.

> ⚠ Note: Any widget API calls will NOT work, as you won't have a valid JWT to send through to gateway. This method is just to remove the need to run all your widgets locally

The hosted widget follows this path pattern:

{baseUrl}/widget/{widgetName}-{widgetVersion}/{default|swimlane}/remoteEntry.js

- baseUrl: (Staff) https://one.sit1.ui.westpac.com.au/ (Customer) https://www.sit1.ui.westpac.com.au/

1. Remove buildManifest.js script from start sequence

./ui/project.json

```
"start": {
  "executor": "nx:run-commands",
  "options": {
    "commands": [
      "node ./tools/envSetup.js",
          "node ./tools/buildManifest.js", // remove this, do not commit this change
      "nx serve coral-banker"
    ],
    "cwd": "ui"
  }
}
```

2. Edit ./ui/public/manifest.json manually

./ui/public/manifest.json

```
{
  ...
  "widgets": [
    {
      "name": "reference-v1",
      "source": "{baseUrl}/widget/{widgetName}-{widgetVersion}/{default|swimlane}/remoteEntry.js"
    }
  ]
}
```

3. Edit you hosts configuration

```
127.0.0.1        local.ui.westpac.com.au
```

4. Stop and restart the MFE

5. Access your SPA via local.ui.westpac.com.au:3000/

Build Errors

403 Not Enough permissions to delete/overwrite artifact

This error maybe be caused if you are trying to:

"rerun failed build/incomplete jobs only" in bamboo pipieline

One of the stages may have published to artifactory and you cannot override an existing artifact that has been published, this is immutable once built.

Solution

1. In Bamboo "Rerun plan" option, do not use "Rerun failed build/incomplete jobs only"

withOptions is not a function

Error: Module build failed (from ../node_modules/postcss-loader/dist/cjs.js)
at /opt/mesh/component/ui/node_modules/@westpac/ui/dist/tailwind/tailwind-plugin.js)

This is an error related to Tailwind when using GEL-Next library.

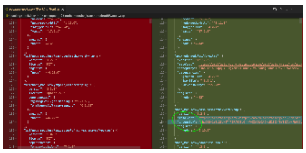
Solution

Please follow instructions managed by GEL-Next

WDP & GEL-Next

npm ERR! Idle timeout reached for host artifactory.srv.westpac.com.au and "cache miss" of packages

This is a result of a package-lock.json missing "resolved" attributes, and overwhelming artifactory by trying to get non-cached versions of packages.
Note: there might be node_modules, package-lock.json files in root, ui, & api folders



Solution

1. `npm cache clean -f`
2. remove all `node_modules` folders
3. remove all `package-lock.json` files
4. `npm install (root project)`
5. Check that "resolved" & "integrity" attributes exist for a majority of packages in package-lock.json files
6. Commit package-lock.json files

npm error code EUSAGE
npm error `npm ci` can only install packages when your package.json and package-lock.json or npm-shrinkwrap.json are in sync.
Please update your lock file with `npm install` before continuing.

if you are using **`latest`** for the Node version in `resources.yaml`, Bamboo will pick up **Node 24** to install packages. This can cause your build to fail with an error similar to:

Build Micro Frontend - indus-homeloanratechangecalculator - Build Artifacts and Container 88: Build…

It's running on node 24 container and build is failing:

12-Dec.-2025 15:52:39	/opt/mesh/component /opt/mesh
12-Dec.-2025 15:52:39	[**INFO**] Running on node version: v24.11.1
12-Dec.-2025 15:52:39	
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "ham-it-up". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "always-auth". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "email". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "ui_ci_npm_registry". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "canvas_binary_host_mirror". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	**npm** warn Unknown global config "tarball". This will stop working in the next major version of npm.
12-Dec.-2025 15:52:39	[**INFO**] Running on npm version: 11.6.2
12-Dec.-2025 15:52:39	
12-Dec.-2025 15:52:39	[**INFO**] Configuring npm permission
12-Dec.-2025 15:52:39	
12-Dec.-2025 15:52:39	[**INFO**] component name: indus-homeloanratechangecalculator
12-Dec.-2025 15:52:39	
12-Dec.-2025 15:52:39	[**INFO**] component type: microFrontend
12-Dec.-2025 15:52:39	

Plain Text

```
npm error code EUSAGE
npm error `npm ci` can only install packages when your package.json and package-lock.json or npm-shrinkwrap.json
are in sync.
Please update your lock file with `npm install` before continuing.
```

```
12-Dec.-2025 15:52:49  npm error code EUSAGE
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error 'npm ci' can only install packages when your package.json and package-lock.json or npm-shrinkwrap.json are in sync. Please update your lock file with 'npm install' before
                       continuing.
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-darwin-arm64@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-darwin-x64@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-linux-arm-gnueabihf@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-linux-arm64-gnu@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-linux-arm64-musl@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-linux-x64-gnu@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-linux-x64-musl@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-win32-arm64-msvc@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @swc/core-win32-ia32-msvc@1.5.29 from lock file
12-Dec.-2025 15:52:49  npm error Missing: fsevents@2.3.3 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/aix-ppc64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/android-arm@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/android-arm64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/android-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/darwin-arm64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/darwin-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/freebsd-arm64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/freebsd-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-arm@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-arm64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-ia32@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-loong64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-mips64el@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-ppc64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-riscv64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-s390x@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/linux-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/netbsd-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/openbsd-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/sunos-x64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/win32-arm64@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @esbuild/win32-ia32@0.20.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: nice-napi@1.0.2 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-darwin-arm64@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-darwin-x64@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-linux-arm-gnueabihf@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-linux-arm64-gnu@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-linux-arm64-musl@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-linux-x64-gnu@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-linux-x64-musl@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error Missing: @nrwl/nx-win32-arm64-msvc@15.9.7 from lock file
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error Clean install a project
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error Usage:
12-Dec.-2025 15:52:49  npm error npm ci
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error Options:
12-Dec.-2025 15:52:49  npm error [--install-strategy <hoisted|nested|shallow|linked>] [--legacy-bundling]
12-Dec.-2025 15:52:49  npm error [--global-style] [--omit <dev|optional|peer> [--omit <dev|optional|peer> ...]]
12-Dec.-2025 15:52:49  npm error [--include <prod|dev|optional|peer> [--include <prod|dev|optional|peer> ...]]
12-Dec.-2025 15:52:49  npm error [--strict-peer-deps] [--foreground-scripts] [--ignore-scripts] [--no-audit]
12-Dec.-2025 15:52:49  npm error [--no-bin-links] [--no-fund] [--dry-run]
12-Dec.-2025 15:52:49  npm error [-w|--workspace <workspace-name> [-w|--workspace <workspace-name> ...]]
12-Dec.-2025 15:52:49  npm error [--workspaces] [--include-workspace-root] [--install-links]
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error aliases: clean-install, ic, install-clean, install-clean
12-Dec.-2025 15:52:49  npm error
12-Dec.-2025 15:52:49  npm error Run "npm help ci" for more info
```

Solution:

- **Option 1**
  Specify the exact Node version you want in `resources.yaml` instead of using `latest`, then trigger the build.
- **Option 2: recommended approach to save future migrations time (If you still want to use `latest`):**
  1. Install Node 24 on your local machine.
  2. Recreate the `package-lock.json` file by running `npm install`.
  3. Push the updated lock file and trigger the build again.

HookWebpackError: ENOENT: no such file or directory

This is a specific error on Windows based machines

Solution

You should set "extractLicenses": false for production configurations for local windows builds to resolve this error
https://bitbucket.srv.westpac.com.au/projects/WDP2/repos/archetype-micro-frontend/browse/templates/ui/project.json#48

ui/project.json

```
"production": {
        "extractLicenses": false
}
```

Deploy Errors

K8s Delpoy - No scopes defined ....

No scope defined for GET

This error will occur when your consumption approval is missing or not correctly scoped.

Solution

Request new consumption grant to downstream API through https://mesh.westpac.com.au/ and ensure that is approved before retrying to deploy your API

K8s Deploy - Could not promote exp/docker

Could not promote exp/docker-exp-<MFE name>-v2:181 at ['wdp-001-docker'] to a0072b-wdp-docker-test

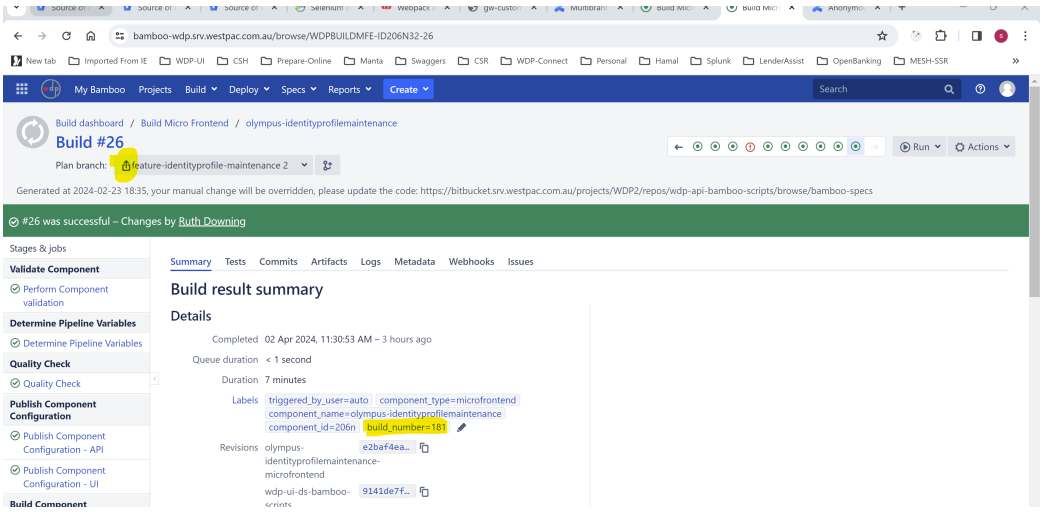
K8s Deploy logs

Custom deployment might have been triggered from a build generated for Pull-request (PR)

When a PR is raised, build would be created for the source branch to ensure if there are any sonar issues on the PR. the build number from this PR shouldn't be promoted to SIT or Dev. Its only for Sonar checks.

How to verify if the build is created from / or for PR.

There will be an icon the build branch, indicating its created from Pull-request



Solution

Always create custom build from master branch to deploy builds from custom branches (Non master), and not rely on builds created with PR icon.

K8s Deploy - Could not deploy exp/docker

error: deployment "{SWIMLANEName}-{SPAName}-xxxx" exceeded its progress deadline

When the docker containers are deployed, there is an automated PIV that requests the /healthcheck endpoint to ensure success.

This check endpoint currently assumes the **fixed port number of 8080 for servicePort**

If your deployments are failing due to the above error,
Do not change the servicePort in app-config.default.json when running a deployment

api/env/default/app-config.default.json

```
{
  "servicePort": "8080",
}
```

Solution

Ensure servicePort is defined as 8080 for platform health checks on deployment can function

If you need change servicePort for local development please update the value inside local config instead api/env/local/app-config.json

Internal GW - Upsert Swagger File Stage

If you are getting messages connecting to "https://apimgr-digital.dev1.api.srv.westpac.com.au/api.." This will indicate that there has been some connectivity to Axway that is causing the deployment failure

Solution

Rerun the build, once and if you experience the same error, please log a ticket platform support

Deploy UI Stage

Failures on this stage are due to missing app-config.json for each respective environment or AWS artifactory not syncing in timely manner with Enterprise Artifactory

Check the following:

Confirm build artifact exists in AWS Artifactory.

SPA

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/singlepageapplication
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/singlepageapplication

MFE

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/microfrontend
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/microfrontend

Widget

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/widget
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/widget

[**ERROR**] Artifact is NOT synced in AWS Artifactory yet

Build artifact is missing in AWS Artifactory. Build files are under "/<SPAName>/build/<buildNumber>/"

Solution

> ⚠ WDP/Mesh have no way of manually forcing a sync event to AWS artifactory. The syncing is managed internally by Enterprise Artifactory
> If this issue persists for more than 60min please raise a service desk ticket with Enterprise DevOps directly https://jira.srv.westpac.com.au/servicedesk/customer/portal/4

1. If there is no artifact above
    a. Wait for your build number artifact to exist in the respective AWS Artifactory above.

2. Once the artifact appears, re-run the respective Mesh deploy pipeline

[**ERROR**] ENV artifact is NOT synced in AWS Artifactory yet

No respective app-config.json for intended deployment environment. ENV specific files are under "/<SPAName>/env/<buildNumber>/<environment>"

Solution

1. If there is no artifact above
    a. Ensure you have a respective app-config.json for each environment you are deploying to in your SPA codebase under "/env /<environment>/"
    b. Wait for your build number artifact to exist in the respective AWS Artifactory above.
2. Once the artifact appears, re-run the respective Mesh deploy pipeline

PIV UI Deployment Stage

Failures on this stage are due to invalid index.html build stamping or AWS Jenkins unpacking pipeline issues

Check the following:

Verify index.html contains the build number stamped as meta tag

SPA

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/singlepageapplication
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/singlepageapplication

MFE

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/microfrontend
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/microfrontend

Widget

Staff: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www1/au/com/westpacgroup/widget
Customer: https://aws-artifactory1.wdp.srv.westpac.com.au/artifactory/webapp/#/artifacts/browse/tree/General/wdp-ui-www/au/com/westpacgroup/widget

a) Download the tgz for the respective build number from artifactory locations above, under "/<SPAName>/env/<buildNumber>/<env>"
b) Extract the files and inspect the index.html source
c) Verify build-number meta tag value in index.html matches the build number downloaded from artifactory.

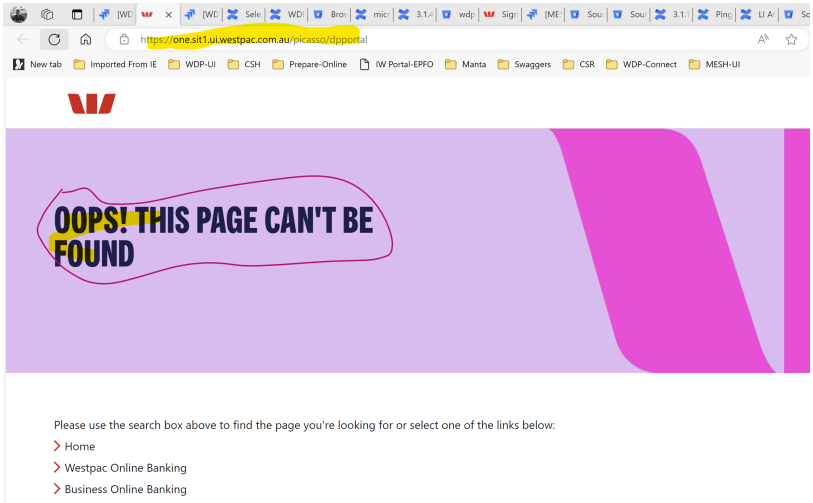
Solution

1. If the build numbers do not match or the file is not present.
    a. Raise a service desk ticket
2. If there is a matching artifact to build number.
    a. Re-run the respective Mesh deploy pipeline
    b. If PIV still fails. This is. AWS Jenkins failure. Raise a service desk ticket

Runtime Errors

Application not accessible / SPA not found

Oops the page cant be found error., when hitting the application URL



Solution 1 - Missing userCommunities

The SPA/MFE will be deployed to hosting environments based on what is declared SPA userCommunities component.yaml eg. https://bitbucket.srv.westpac.com.au/projects/WDP/repos/showcase-singlepageapplication/browse/component.yaml#18

- customer will be hosted at http://**www**.<env>.ui.westpac.com.au
- staff will be hosted at http://**one**..ui.westpac.com.au

If these are not defined, they will NOT be deployed to the hosted environments

Solution 2 - CSP is not configured

If you have tried solution 1 and have confirmed userCommunities is defined, and you are on correct url host. Your CSP might be missing setting to enable traffic to flow to singlePageApplication
`"singlePageApplication": true`

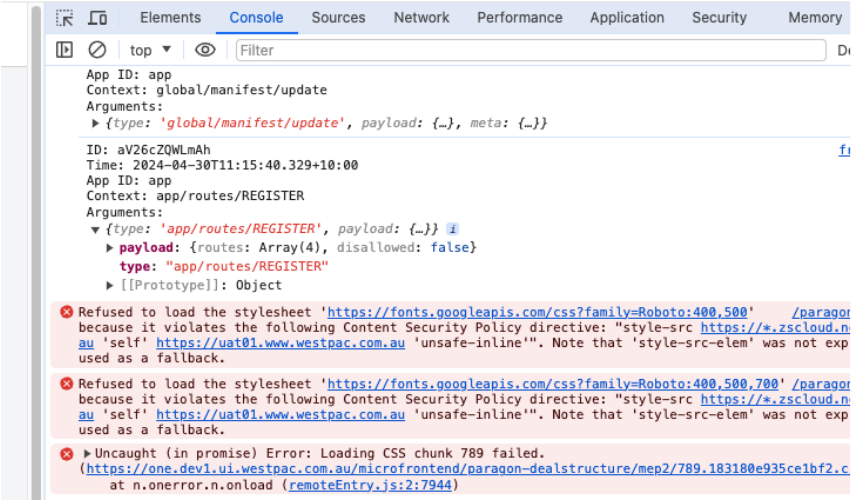
For more details refer 3.1.3 Deploy SPA and MFE - Westpac Mesh - Westpac Enterprise Confluence

Solution 3 - App-config is too large for stamping

If the deploy has succeeded, but browsing to the index.html file results in a 404 error, it is possibly related to the app using DTM Launch capability, and the auto stamping of app-config.json contents exceeding the limits at Cloudfront. Refer to App config stamping to increase performance in MFE

Uncaught (in promise) Error: Loading chunk ...

If remoteEntry.js (MFE entry file) fails to load any dependency javascipt or css files, (This will usually be shown in the network tab or console as failures) the MFE will not render, and will show the loader by default.



This can be due either:

1. Files not being deployed in MFE, eg. 404 errors or request is returning default 404 error page
2. Files being blocked by Content Security Policy eg. console or network tab refers to "violates Content Security Policy directive"

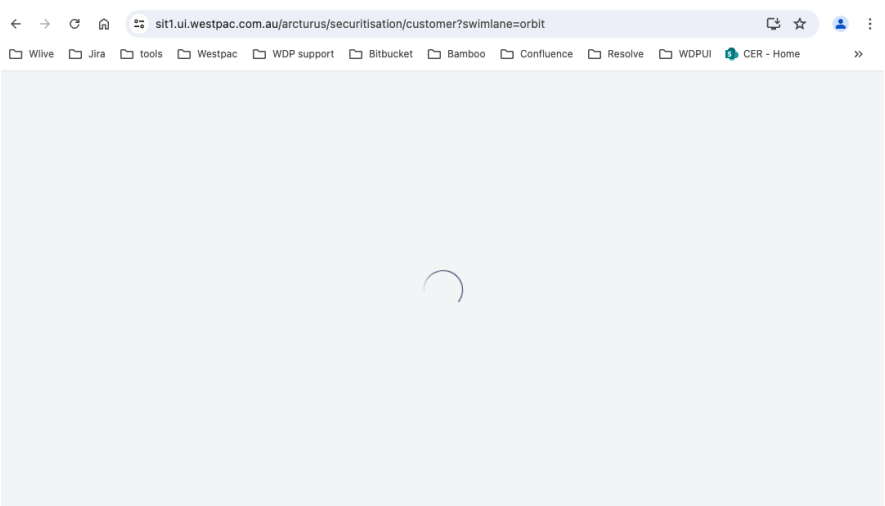
Solution - Missing javascript files

Redeploy your MFE, and validate the file is returning javascript

Solution - Content Security Policy (CSP) block

Ensure any blocked files are intended to be loaded from the public internet (Please consult your SBD consultant). If appropriate with SBD, whitelist the hostname to the blocked resource in your relevant CSP settings https://bitbucket.srv.westpac.com.au/projects/UI-SEC

Application loader keeps on loading and routes disallowed



Check on the console, if MFE routes are registered successfully, if you see disallowed as true, then logged in user dont have valid roles to access application or config issue.

```
framework_core.154f4239.chunk.js:1
ID: CE1HRBOI-G5V
Time: 2024-07-02T14:18:15.579+10:00
App ID: app
Context: app/routes/REGISTER
Arguments:

▼ {type: 'app/routes/REGISTER', payload: {…}} i
  ▼ payload:
      disallowed: true
    ▼ routes: Array(5)
      ▶ 0: {id: 'pages.arcturus.securitisation.home', path: '/securitisation/*', isPublic: false, name: 'Arcturus-Securitisation', element: {…}, …}
      ▶ 1: {id: 'pages.arcturus.securitisation.homepage', index: true, isPublic: false, name: 'Arcturus-Securitisation', roles: Array(1), …}
      ▶ 2: {id: 'pages.arcturus.securitisation.pageone', path: 'page-one', isPublic: false, name: 'Arcturus-Securitisation/page-one', roles: Array(1)
      ▶ 3: {id: 'pages.arcturus.securitisaton.activecustomer', path: 'customer', isPublic: false, name: 'Arcturus-Securitisation/customer', roles: Ar
      ▶ 4: {id: 'pages.arcturus.securitisation.error', name: 'Arcturus-Securitisation - Error', path: '*', isPublic: true, element: {…}, …}
        length: 5
      ▶ [[Prototype]]: Array(0)
    ▶ [[Prototype]]: Object
    type: "app/routes/REGISTER"
  ▶ [[Prototype]]: Object
```

If the logged in user got correct roles, but still got above error fix by following below.

1. Check for user role in id_token by copy pasting the id_token in https://jwt.io and access_token.


```
▼ {,…}
    access_token: "eyJraWQiOiJOVDI4a0ZWSk5IS2cxclg0ZnFDdlpjRkhwbjRvVEZXRlBKNEhmNXVNWlhBIiwiYWxnIjoiUlMyNTYifQ.eyJzdWIiOiJudWxsIi
    expires_at: 1719894618
    id_token: "eyJraWQiOiJOVDI4a0ZWSk5IS2cxclg0ZnFDdlpjRkhwbjRvVEZXRlBKNEhmNXVNWlhBIiwiYWxnIjoiUlMyNTYifQ.eyJ1c2VySWQiOiIwMDc4ND
  ▶ profile: {userId: "00784020025", rt_hash: "MIG4wh5ckbN-wxvNIp7olA", brandsilo: "WPAC",…}
    refresh_token: "stLjFVFX9GJivI5NpTJKmnswEprjj5zwPv3uujrt"
    scope: "openid arcturus"
    token_type: "bearer"
```


2. If role is not available in id_token and present only in access_token, then update oidc config in spa with the below flag "shouldSpreadAccessToken": true

503 error returned from BFF API

If you are getting sporadic / random 503 errors to BFF API check following points to identify any code issues.

Possible RCA & Solution

Check the node version you are using. From node 20, if there is any unhandled promise rejection it would crash the container and subsequent calls to BFF API would result in 503 error, until fresh deployment happens.

Its behaviour of node from 20 version, Before node 20, it use to warn there is unhandled promise rejection, whereas from node 20 it would crash the container.

Debug steps

- Upgrade local to use node20 and should be able able to replicate in local.
- Validate on the code, if there are unhandled promise rejection.
- If any promises are called as fire and forget ensure (Even Kafka audit calls), it has explicit catch block. Promise without async / await even composed within try-catch would lead to unhandled exception.
- Debug on 503 correlation may not lead to actual error, need to look for last 500 / 200 errors if it had any exception which could have crashed container.

504 error returned from gateway

The default timeout set at platform gateway is 60s

resources.yaml

```
http:
  requestTimeoutSeconds: 120
```

500 error timeout returned from BFF API

The default timeout value for node based BFF/EXP API is 15s

app-config.json

```
{
  "timeouts": {
    "/your/resource/path": 30000 // in milliseconds, ensure your gateway timeout (see above) is greater than
this value
  }
}
```

High CPU observed in containers

A recent update to platform NodeJs containers to enable OTEL observability metrics, has seen an uptick in CPU.

For 2.26 platform release, OTEL will be defaulted to off ie. OTEL_DISABLED=true

If you require to have OTEL capability please deploy your MFE with the custom build flag

```
OTEL_DISABLED=false
```

It is recommended you use the OTEL_DISABLED flag until the OTEL issue is resolved in Node containers, alternatively, the following can be used instead of the OTEL_DISABLED

resources.yaml

```
observability:
  openTelemetry:
    enabled: true
```

healthcheck returns response code 0

healthcheck endpoints are used by Mesh platform in determining whether to restart containers that are returning non 2xx response codes. eg. 5xx code

As the healthcheck resource is highly used, the gateway allows a very small timeout window for the healthcheck to return a response back to the gateway. **This is currently a timeout of 1 second, and is very specific to the healthcheck resource.** Generally your application specific resource endpoints timeout at 15 seconds.

When you have blocking requests/operations inside the healthcheck resource, that raises the possibility that the healthcheck resource might breech the timeout value the gateway will return a response code of 0. This is expected on the healthcheck endpoint. Examples of blocking requests might be set/get to Redis clusters

> ⚠ **Getting a response code of 0 on heathcheck endpoint, does not indicate that your other resources are unreachable.**

In order to validate if your actual operations endpoints are functional please try the following splunk query

```
MFE framework:


index=prd_a008e0 sourcetype="startup-error:container:wdp" | rex field=log "@mep-node/framework-cache-provider-
addon::error RedisProvider… :: an error occurred (?<error_message>.*)" | rex field=source "/logs/(?<component>[^
/]+)/"
| search error_message=* component=spica-transactionstriage
| table _time, component, error_message
| sort component _time

Legacy Exp framework:
index=prd_a008e0 source="/logs/cap-technolmgt-applnmgt-uiutils-v1/*/*.json" "Redis :: Error connecting to redis
:: host " | table _time component_name msg | sort - _time
```